Data Cleaning Techniques for Banking Professionals

All-in-one PDF with concepts, use cases, examples, mini datasets, and ready-to-copy prompts for non-technical participants.

Raw Data → Cleaning Rule → AI Prompt → Human Review → MIS / Workflow

Who is this for?

Banking operations, branch staff, compliance users, customer support, process teams, analysts, and non-technical managers.

How to use this PDF

Participants can copy any prompt directly from this PDF and run it in ChatGPT/Gemini. No separate dataset file is required because each prompt includes a mini dataset.

What is included?

25 data cleaning techniques. Each technique includes a plain-English explanation, banking use case, sample data, and a ready-to-copy prompt.

Safety note

Use dummy or masked data during training. Never paste real account numbers, NRIC/passport details, card numbers, passwords, or confidential bank data into public AI tools.

Practice Flow for Participants

Read issue → Copy prompt → Run AI → Check output → Decide action

Step	Participant action	Trainer note
1	Choose one cleaning technique	Explain the business problem first, not the technical method.
2	Copy the full prompt box	Each prompt already includes a mini dataset.
3	Paste into ChatGPT/Gemini	Ask participants to review the output like banking staff.
4	Discuss human validation	AI suggests; banking staff verifies.
5	Connect to MIS/workflow	Explain how clean data improves reports, compliance, and customer service.

Technique Index

No	Technique
01	Remove Duplicate Records
02	Handle Missing Values
03	Standardize Date Formats
04	Standardize Customer Names
05	Standardize Branch and City Names
06	Convert Amounts to Numbers
07	Remove Extra Spaces and Fix Case
08	Validate Account Numbers
09	Identify Outliers
10	Remove Invalid Characters
11	Standardize Product Names
12	Clean Complaint Text
13	Standardize Yes/No Fields

No	Technique
14	Check Category Mapping
15	Validate Email and Mobile Numbers
16	Remove Test or Dummy Records
17	Split One Column into Many
18	Combine Columns for Communication
19	Address Cleaning
20	Currency Standardization
21	Remove Old or Inactive Records
22	Check Data Freshness
23	Reconcile Totals Across Sources
24	Clean Headers and Column Names
25	Create Error Flags Instead of Deleting

01. Remove Duplicate Records

2

Simple explanation

Same customer, account, or complaint appears more than once.

Banking use case

Prevents inflated customer counts, repeated SMS/email, duplicate KYC checks, and repeated complaint counting.

Sample mini dataset

Record_ID	Customer_ID	Account_No	Email	Mobile	Complaint_Text
R001	C1001	100045678901	ahmad@email.com	60123450001	ATM failed but debit happened
R002	C1001	100045678901	ahmad@email.com	60123450001	ATM failed but debit happened
R003	C1002	100045678902	siti@email.com	60123450002	Card not delivered

Trainer line

"Data cleaning is a business quality activity. For this technique, participants should focus on the risk of using the data without checking it."

READY-TO-COPY PARTICIPANT PROMPT

```
You are a banking data quality reviewer.

Analyze this mini dataset and identify possible duplicate records.
Return a table with: Record_ID, Duplicate_Group, Duplicate_Reason,
Confidence, Recommended_Action, Human_Check_Needed.

Business rule: Mark as strong duplicate only when at least two
identifiers match, such as Customer_ID + Account_No or Email + Mobile.
Do not delete anything automatically.

Dataset:
Record_ID | Customer_ID | Account_No | Email | Mobile | Complaint_Text
R001 | C1001 | 100045678901 | ahmad@email.com | 60123450001 | ATM
failed but debit happened
R002 | C1001 | 100045678901 | ahmad@email.com | 60123450001 | ATM
failed but debit happened
R003 | C1002 | 100045678902 | siti@email.com | 60123450002 | Card not
delivered
```

02. Handle Missing Values

Simple explanation

Important fields such as city, branch, KYC status, email, or income are blank.

Banking use case

Helps staff decide whether to fill as Unknown, leave blank, or send for verification.

Sample mini dataset

Record_ID	Customer_Name	City	KYC_Status	Email	Monthly_Income
R011	Ameer Khan		Completed	ameer@email.com	6500
R012	Lim Wei	Penang		lim@email.com	4800
R013	Siti Aminah	Kuala Lumpur	Pending		

Trainer line

"Data cleaning is a business quality activity. For this technique, participants should focus on the risk of using the data without checking it."

READY-TO-COPY PARTICIPANT PROMPT

```
You are a banking data cleaning assistant.

Analyze this mini dataset for missing values.
For each missing value, decide one action: Fill as Unknown, Flag for
Verification, or Leave Blank.
Return a table with: Record_ID, Column_With_Missing_Value, Risk_Level,
Recommended_Action, Reason.

Rules:
- Do not invent missing customer information.
- KYC, customer contact, and income fields should be treated as higher
risk than simple reporting fields.

Dataset:
Record_ID | Customer_Name | City | KYC_Status | Email | Monthly_Income
R011 | Ameer Khan | | Completed | ameer@email.com | 6500
R012 | Lim Wei | Penang | | lim@email.com | 4800
R013 | Siti Aminah | Kuala Lumpur | Pending | |
```

03. Standardize Date Formats

Simple explanation

Dates appear in multiple formats, causing wrong daily or monthly reporting.

Banking use case

Improves MIS reports, failed transaction monitoring, and audit evidence preparation.

Sample mini dataset

Record_ID	Transaction_Date	Last_Updated
R021	24/06/2026	2026-06-25
R022	2026/06/24	25-Jun-2026
R023	06/07/26	07-06-2026

Trainer line

"Data cleaning is a business quality activity. For this technique, participants should focus on the risk of using the data without checking it."

READY-TO-COPY PARTICIPANT PROMPT

```
You are preparing a banking MIS dataset.

Standardize the dates below to YYYY-MM-DD where it is safe to do so.
Flag ambiguous or invalid dates for human review.
Return a table with: Record_ID, Original_Field, Original_Date_Value,
Suggested_Clean_Date, Status, Reason.

Important: Do not guess ambiguous dates such as 06/07/26 unless the
day/month rule is clearly known.

Dataset:
Record_ID | Transaction_Date | Last_Updated
R021 | 24/06/2026 | 2026-06-25
R022 | 2026/06/24 | 25-Jun-2026
R023 | 06/07/26 | 07-06-2026
```

04. Standardize Customer Names

Simple explanation

Names may have uppercase, lowercase, initials, symbols, or extra spaces.

Banking use case

Helps customer matching, KYC review, letter generation, and complaint analysis.

Sample mini dataset

Record_ID	Customer_ID	Customer_Name	Account_No
R031	C3001	AHMAD RAHMAN	100000000001
R032	C3002	siti aminah	100000000002
R033	C3003	M. Ali##	100000000003

Trainer line

"Data cleaning is a business quality activity. For this technique, participants should focus on the risk of using the data without checking it."

READY-TO-COPY PARTICIPANT PROMPT

```
You are a banking customer master data reviewer.

Standardize customer names into clean proper case. Remove extra spaces
and unnecessary symbols.
Do not merge customers based only on similar names.
Return a table with: Record_ID, Original_Name, Suggested_Clean_Name,
Possible_Issue, Human_Check_Needed.

Dataset:
Record_ID | Customer_ID | Customer_Name | Account_No
R031 | C3001 | AHMAD RAHMAN | 100000000001
R032 | C3002 | siti aminah | 100000000002
R033 | C3003 | M. Ali## | 100000000003
```

05. Standardize Branch and City Names

Simple explanation

Branch and city names may be entered as short forms, typos, or mixed formats.

Banking use case

Improves branch performance reports and city-wise customer dashboards.

Sample mini dataset

Record_ID	Branch	City
R041	KL Main	Kualalumpur
R042	K.L. Main Branch	Kuala Lumpur
R043	Penang Georgetown	George Town

Trainer line

"Data cleaning is a business quality activity. For this technique, participants should focus on the risk of using the data without checking it."

READY-TO-COPY PARTICIPANT PROMPT

```
You are cleaning branch and city data for a Malaysian banking report.

Find inconsistent spellings or short forms and suggest standard values.
Return a table with: Record_ID, Field_Name, Original_Value,
Suggested_Standard_Value, Confidence, Reason.

Rule: One real branch/city should appear as one standard value in the
final MIS report.

Dataset:
Record_ID | Branch | City
R041 | KL Main | Kualalumpur
R042 | K.L. Main Branch | Kuala Lumpur
R043 | Penang Georgetown | George Town
```

06. Convert Amounts to Numbers

Simple explanation

Amounts may contain RM symbols, commas, text, or slash notation.

Banking use case

Needed before calculating totals, averages, dashboards, and risk thresholds.

Sample mini dataset

Record_ID	Amount	Currency
R051	RM 2,500	MYR
R052	3,750.00	MYR
R053	Five Thousand	MYR
R054	2500/-	MYR

Trainer line

"Data cleaning is a business quality activity. For this technique, participants should focus on the risk of using the data without checking it."

READY-TO-COPY PARTICIPANT PROMPT

```
You are a banking report preparation assistant.

Identify amounts stored as text or with symbols. Suggest clean numeric
values where safe.
Return a table with: Record_ID, Original_Amount,
Suggested_Numeric_Amount, Currency, Status, Reason.

Rules:
- Amount should be numeric.
- Currency should remain separate.
- Do not guess values written unclearly in words.

Dataset:
Record_ID | Amount | Currency
R051 | RM 2,500 | MYR
R052 | 3,750.00 | MYR
R053 | Five Thousand | MYR
R054 | 2500/- | MYR
```

07. Remove Extra Spaces and Fix Case

Simple explanation

Values look similar but differ due to spaces or capitalization.

Banking use case

Prevents hidden duplicates in branch, product, city, and status reports.

Sample mini dataset

Record_ID	Customer_Name	Branch	Product	KYC_Status
R061	Raj Kumar	kl main	credit card	completed
R062	MEI LING	PENANG	SAVINGS ACCOUNT	PENDING
R063	Siti Aminah	Kuala Lumpur Main	Debit Card	Done

Trainer line

"Data cleaning is a business quality activity. For this technique, participants should focus on the risk of using the data without checking it."

READY-TO-COPY PARTICIPANT PROMPT

```
You are cleaning text fields for a banking dashboard.

Remove leading/trailing spaces, replace multiple spaces with one space,
and standardize capitalization.
Return a table with: Record_ID, Column_Name, Original_Value,
Cleaned_Value, Why_Changed.

Do not change the meaning of any value.

Dataset:
Record_ID | Customer_Name | Branch | Product | KYC_Status
R061 | Raj Kumar | kl main | credit card | completed
R062 | MEI LING | PENANG | SAVINGS ACCOUNT | PENDING
R063 | Siti Aminah | Kuala Lumpur Main | Debit Card | Done
```

08. Validate Account Numbers

Simple explanation

Account numbers may have letters, wrong length, or symbols.

Banking use case

Reduces failed transactions and customer mapping errors before bulk processing.

Sample mini dataset

Record_ID	Account_No
R071	100045678901
R072	ABC123456
R073	12345
R074	1000-4567-8901

Trainer line

"Data cleaning is a business quality activity. For this technique, participants should focus on the risk of using the data without checking it."

READY-TO-COPY PARTICIPANT PROMPT

```
You are a banking operations control reviewer.

Validate the Account_No values using this training rule:
- Account_No must contain only digits.
- Account_No should be between 10 and 14 digits.

Return a table with: Record_ID, Account_No, Validation_Status,
Issue_Found, Recommended_Action.

Do not create or guess a corrected account number.

Dataset:
Record_ID | Account_No
R071 | 100045678901
R072 | ABC123456
R073 | 12345
R074 | 1000-4567-8901
```

09. Identify Outliers

Simple explanation

Some amounts are far higher/lower than normal and need review.

Banking use case

Supports fraud review, AML monitoring, and operational error detection.

Sample mini dataset

Record_ID	Customer_Segment	Transaction_Type	Amount
R081	Retail	ATM Withdrawal	300
R082	Retail	ATM Withdrawal	250
R083	Retail	ATM Withdrawal	950000
R084	SME	Online Transfer	12000

Trainer line

"Data cleaning is a business quality activity. For this technique, participants should focus on the risk of using the data without checking it."

READY-TO-COPY PARTICIPANT PROMPT

```
You are a banking fraud and operations reviewer.

Identify unusual transaction amounts in this mini dataset.
Return a table with: Record_ID, Amount, Why_Unusual,
Possible_Explanation, Recommended_Action, Human_Check_Needed.

Important: Do not label an outlier as fraud automatically. Mark it for
review.

Dataset:
Record_ID | Customer_Segment | Transaction_Type | Amount
R081 | Retail | ATM Withdrawal | 300
R082 | Retail | ATM Withdrawal | 250
R083 | Retail | ATM Withdrawal | 950000
R084 | SME | Online Transfer | 12000
```

10. Remove Invalid Characters

Simple explanation

Names and addresses may include symbols that affect letters, statements, or matching.

Banking use case

Protects professionalism in customer communication and document generation.

Sample mini dataset

Record_ID	Customer_Name	Address
R091	Ravi Kumar###	No. 12, Jalan Ampang!!!
R092	Siti@Aminah	##22 Jalan Tun Razak
R093	John Tan	15, Lebuh Pantai

Trainer line

"Data cleaning is a business quality activity. For this technique, participants should focus on the risk of using the data without checking it."

READY-TO-COPY PARTICIPANT PROMPT

```
You are cleaning customer communication data.

Identify unnecessary symbols in Customer_Name and Address. Suggest
cleaned values.
Return a table with: Record_ID, Field_Name, Original_Value,
Suggested_Clean_Value, Risk_If_Not_Cleaned.

Rules:
- Preserve meaningful address punctuation where useful.
- Do not remove information that may be part of a valid address.

Dataset:
Record_ID | Customer_Name | Address
R091 | Ravi Kumar### | No. 12, Jalan Ampang!!!
R092 | Siti@Aminah | ##22 Jalan Tun Razak
R093 | John Tan | 15, Lebuh Pantai
```

11. Standardize Product Names

Simple explanation

Products may be written as abbreviations, short forms, or variations.

Banking use case

Improves product-level complaint analysis and management dashboards.

Sample mini dataset

Record_ID	Product	Complaint_Text
R101	CC	Card payment failed
R102	Credit-Card	Late fee dispute
R103	Cr Card	Card not delivered
R104	Savings Acct	Statement issue

Trainer line

"Data cleaning is a business quality activity. For this technique, participants should focus on the risk of using the data without checking it."

READY-TO-COPY PARTICIPANT PROMPT

```
You are cleaning product names for a banking complaint dashboard.

Map product variations to standard product categories.
Return a table with: Record_ID, Original_Product,
Suggested_Standard_Product, Reason, Confidence.

Suggested categories: Credit Card, Debit Card, Savings Account, Current
Account, Loan, Online Banking, ATM.

Dataset:
Record_ID | Product | Complaint_Text
R101 | CC | Card payment failed
R102 | Credit-Card | Late fee dispute
R103 | Cr Card | Card not delivered
R104 | Savings Acct | Statement issue
```

12. Clean Complaint Text

Simple explanation

Complaints may contain emotional, repeated, or unclear text.

Banking use case

Helps classify complaints and route them to the right department.

Sample mini dataset

Record_ID	Complaint_Text
R111	Payment failed failed failed!!! Amount deducted. Urgent urgent.
R112	No response from support. I called 3 times.
R113	ATM cash not dispensed but account debited.

Trainer line

"Data cleaning is a business quality activity. For this technique, participants should focus on the risk of using the data without checking it."

READY-TO-COPY PARTICIPANT PROMPT

```
You are a banking complaint cleaning assistant.

Clean and summarize the complaint text without changing the meaning.
Return a table with: Record_ID, Cleaned_Complaint, Short_Summary,
Complaint_Category, Urgency_Level.

Rules:
- Remove repeated words and unnecessary punctuation.
- Preserve customer issue and urgency.
- Do not invent details.

Dataset:
Record_ID | Complaint_Text
R111 | Payment failed failed failed!!! Amount deducted. Urgent urgent.
R112 | No response from support. I called 3 times.
R113 | ATM cash not dispensed but account debited.
```

13. Standardize Yes/No Fields

Simple explanation

Yes/No fields may be written as Y, N, Done, Completed, Pending, or blank.

Banking use case

Improves KYC, consent, and compliance completion reports.

Sample mini dataset

Record_ID	KYC_Completed	Marketing_Consent
R121	Y	Yes
R122	Completed	N
R123	Done	No
R124	Pending	

Trainer line

"Data cleaning is a business quality activity. For this technique, participants should focus on the risk of using the data without checking it."

READY-TO-COPY PARTICIPANT PROMPT

```
You are cleaning Yes/No and status fields for a compliance report.

Standardize values into: Yes, No, Pending, or Review Required.
Return a table with: Record_ID, Field_Name, Original_Value,
Standard_Value, Reason.

Do not treat blank as Yes.

Dataset:
Record_ID | KYC_Completed | Marketing_Consent
R121 | Y | Yes
R122 | Completed | N
R123 | Done | No
R124 | Pending |
```

14. Check Category Mapping

Simple explanation

Transaction descriptions may be mapped to the wrong category.

Banking use case

Improves customer spend analysis, fraud review, and operational reporting.

Sample mini dataset

Record_ID	Description	Assigned_Category
R131	ATM Withdrawal at KLCC	Online Transfer
R132	Loan EMI payment	Shopping
R133	Credit card bill payment	Credit Card Payment

Trainer line

"Data cleaning is a business quality activity. For this technique, participants should focus on the risk of using the data without checking it."

READY-TO-COPY PARTICIPANT PROMPT

```
You are reviewing transaction category mapping.

Compare Description and Assigned_Category. Identify possible mismatches
and suggest corrected categories.
Return a table with: Record_ID, Description, Assigned_Category,
Suggested_Category, Issue_Found, Confidence.

Dataset:
Record_ID | Description | Assigned_Category
R131 | ATM Withdrawal at KLCC | Online Transfer
R132 | Loan EMI payment | Shopping
R133 | Credit card bill payment | Credit Card Payment
```

15. Validate Email and Mobile Numbers

Simple explanation

Invalid contact details prevent OTP, alerts, statements, and notices.

Banking use case

Reduces customer service issues and failed notifications.

Sample mini dataset

Record_ID	Email	Mobile
R141	ahmad@gmail.com	60123456789
R142	siti.gmail.com	12345
R143	raj@	60 17 888 9999
R144		60199887766

Trainer line

"Data cleaning is a business quality activity. For this technique, participants should focus on the risk of using the data without checking it."

READY-TO-COPY PARTICIPANT PROMPT

```
You are validating customer contact details for a bank.

Check Email and Mobile values and flag invalid or incomplete records.
Return a table with: Record_ID, Field_Name, Original_Value,
Validation_Status, Issue_Found, Recommended_Action.

Rules:
- Do not guess missing emails or mobile numbers.
- For training, assume Malaysian mobile numbers should be in a clear
format beginning with 60 and should contain enough digits for a valid
contact number.

Dataset:
Record_ID | Email | Mobile
R141 | ahmad@gmail.com | 60123456789
R142 | siti.gmail.com | 12345
R143 | raj@ | 60 17 888 9999
R144 | | 60199887766
```

16. Remove Test or Dummy Records

Simple explanation

Training/test records can be mixed with production reports.

Banking use case

Prevents wrong customer growth, branch performance, and campaign reports.

Sample mini dataset

Record_ID	Customer_Name	Account_No	Remarks
R151	Test User	0000000000	system test
R152	Dummy Customer	1234567890	training entry
R153	Nur Aisyah	100078945612	real customer

Trainer line

"Data cleaning is a business quality activity. For this technique, participants should focus on the risk of using the data without checking it."

READY-TO-COPY PARTICIPANT PROMPT

```
You are preparing a clean customer report.

Identify likely test or dummy records.
Return a table with: Record_ID, Customer_Name, Reason_Flagged,
Recommended_Action, Human_Check_Needed.

Rules:
- Do not remove records automatically.
- Flag suspicious test/dummy records for business confirmation.

Dataset:
Record_ID | Customer_Name | Account_No | Remarks
R151 | Test User | 0000000000 | system test
R152 | Dummy Customer | 1234567890 | training entry
R153 | Nur Aisyah | 100078945612 | real customer
```

17. Split One Column into Many

Simple explanation

One field may contain name, city, segment, or product together.

Banking use case

Makes filtering, grouping, segmentation, and dashboard building easier.

Sample mini dataset

Record_ID	Combined_Field
R161	Ahmad Rahman - Kuala Lumpur - Premium
R162	Siti Aminah - Penang - Mass
R163	Raj Kumar - Johor Bahru - SME

Trainer line

"Data cleaning is a business quality activity. For this technique, participants should focus on the risk of using the data without checking it."

READY-TO-COPY PARTICIPANT PROMPT

```
You are restructuring customer data for reporting.

Split the Combined_Field into Customer_Name, City, and Segment.
Return a table with: Record_ID, Customer_Name, City, Segment,
Issue_Found.

Rule: If the field cannot be split confidently, mark it as Review
Required.

Dataset:
Record_ID | Combined_Field
R161 | Ahmad Rahman - Kuala Lumpur - Premium
R162 | Siti Aminah - Penang - Mass
R163 | Raj Kumar - Johor Bahru - SME
```

18. Combine Columns for Communication

Simple explanation

Letters and messages often need full names or full addresses.

Banking use case

Improves statement generation, letters, and customer notifications.

Sample mini dataset

Record_ID	First_Name	Last_Name	Address_Line1	City
R171	Siti	Aminah	12 Jalan Ampang	Kuala Lumpur
R172	Lim	Wei	88 Lebuh Pantai	Penang

Trainer line

"Data cleaning is a business quality activity. For this technique, participants should focus on the risk of using the data without checking it."

READY-TO-COPY PARTICIPANT PROMPT

```
You are preparing customer communication fields.

Combine First_Name and Last_Name into Full_Name. Combine Address_Line1
and City into Mailing_Address.
Return a table with: Record_ID, Full_Name, Mailing_Address,
Issue_Found.

Rules:
- Do not invent missing address details.
- Keep spacing professional.

Dataset:
Record_ID | First_Name | Last_Name | Address_Line1 | City
R171 | Siti | Aminah | 12 Jalan Ampang | Kuala Lumpur
R172 | Lim | Wei | 88 Lebuh Pantai | Penang
```

19. Address Cleaning

Simple explanation

Addresses may use abbreviations, missing punctuation, or inconsistent city names.

Banking use case

Supports KYC, statement delivery, branch communication, and audit checks.

Sample mini dataset

Record_ID	Address
R181	No 12 Jln Ampang KL
R182	#22, Jalan Tun Razak, K.L.
R183	88 Lebuh Pantai Penang

Trainer line

"Data cleaning is a business quality activity. For this technique, participants should focus on the risk of using the data without checking it."

READY-TO-COPY PARTICIPANT PROMPT

```
You are cleaning address data for a banking KYC review.

Suggest cleaner address formats and flag unclear addresses.
Return a table with: Record_ID, Original_Address,
Suggested_Clean_Address, Issue_Found, Human_Check_Needed.

Rules:
- Do not invent missing house numbers or postal codes.
- Expand obvious abbreviations only when safe.

Dataset:
Record_ID | Address
R181 | No 12 Jln Ampang KL
R182 | #22, Jalan Tun Razak, K.L.
R183 | 88 Lebuh Pantai Penang
```

20. Currency Standardization

Simple explanation

Amount and currency may be mixed or missing.

Banking use case

Essential for remittance, treasury, card settlement, and international reports.

Sample mini dataset

Record_ID	Amount_Text
R191	RM 1,000
R192	USD 1,000
R193	SGD1000
R194	1000

Trainer line

"Data cleaning is a business quality activity. For this technique, participants should focus on the risk of using the data without checking it."

READY-TO-COPY PARTICIPANT PROMPT

```
You are cleaning multi-currency transaction data.

Separate amount and currency where possible. Flag records where
currency is missing.
Return a table with: Record_ID, Original_Amount_Text, Suggested_Amount,
Suggested_Currency, Status, Reason.

Important: RM 1,000 and USD 1,000 are not the same business value. Do
not assume currency when missing.

Dataset:
Record_ID | Amount_Text
R191 | RM 1,000
R192 | USD 1,000
R193 | SGD1000
R194 | 1000
```

21. Remove Old or Inactive Records

Simple explanation

Closed/dormant accounts may appear in active customer reports.

Banking use case

Ensures reports include the right customer population.

Sample mini dataset

Record_ID	Customer_Name	Account_Status	Last_Transaction_Date
R201	Ameer Khan	Active	2026-06-24
R202	Lim Wei	Closed	2023-01-10
R203	Mei Ling	Dormant	2021-05-01

Trainer line

"Data cleaning is a business quality activity. For this technique, participants should focus on the risk of using the data without checking it."

READY-TO-COPY PARTICIPANT PROMPT

```
You are preparing an active customer report.

Identify records that should be excluded or flagged based on
Account_Status and Last_Transaction_Date.
Return a table with: Record_ID, Account_Status, Last_Transaction_Date,
Include_In_Active_Report, Reason.

Rules:
- Active report should focus on active customers.
- Closed and dormant records may be kept in historical reports but not
active customer counts.

Dataset:
Record_ID | Customer_Name | Account_Status | Last_Transaction_Date
R201 | Ameer Khan | Active | 2026-06-24
R202 | Lim Wei | Closed | 2023-01-10
R203 | Mei Ling | Dormant | 2021-05-01
```

22. Check Data Freshness

12

Simple explanation

Correct data can still be risky if it is outdated.

Banking use case

Useful before fraud alerts, contact campaigns, KYC refresh, and customer notifications.

Sample mini dataset

Record_ID	Email	Mobile	Last_Updated
R211	ahmad@email.com	60123450001	2026-06-20
R212	siti@email.com	60123450002	2024-01-05
R213	lim@email.com	60123450003	2022-11-15

Trainer line

"Data cleaning is a business quality activity. For this technique, participants should focus on the risk of using the data without checking it."

READY-TO-COPY PARTICIPANT PROMPT

```
You are checking customer contact data freshness.

Assume today is 2026-06-26 for this training exercise.
Flag records where Last_Updated is older than 12 months.
Return a table with: Record_ID, Last_Updated, Freshness_Status, Risk,
Recommended_Action.

Dataset:
Record_ID | Email | Mobile | Last_Updated
R211 | ahmad@email.com | 60123450001 | 2026-06-20
R212 | siti@email.com | 60123450002 | 2024-01-05
R213 | lim@email.com | 60123450003 | 2022-11-15
```

23. Reconcile Totals Across Sources

Simple explanation

Core banking, MIS, and manual spreadsheets may show different totals.

Banking use case

Prevents submitting wrong reports to management or regulators.

Sample mini dataset

Source	Transaction_Count	Total_Amount
Core Banking	10000	12500000
MIS Report	9850	12350000
Manual Excel	10000	12500000

Trainer line

"Data cleaning is a business quality activity. For this technique, participants should focus on the risk of using the data without checking it."

READY-TO-COPY PARTICIPANT PROMPT

```
You are a banking MIS reconciliation reviewer.

Compare the totals across sources and identify mismatches.
Return a table with: Compared_Sources, Difference_In_Count,
Difference_In_Amount, Risk_Level, Recommended_Action.

Also provide a short explanation of why reconciliation is important
before submitting reports.

Dataset:
Source | Transaction_Count | Total_Amount
Core Banking | 10000 | 12500000
MIS Report | 9850 | 12350000
Manual Excel | 10000 | 12500000
```

24. Clean Headers and Column Names

Simple explanation

Column names may be short, unclear, duplicated, or inconsistent.

Banking use case

Helps business users, dashboards, and AI tools understand data correctly.

Sample mini dataset

Raw_Header	Suggested_Header
Cust No	Customer_ID
Txn Amt	Transaction_Amount
Br	Branch
Stat	Status

Trainer line

"Data cleaning is a business quality activity. For this technique, participants should focus on the risk of using the data without checking it."

READY-TO-COPY PARTICIPANT PROMPT

```
You are cleaning column headers for a banking dataset.

Review the raw column headers and suggest clear business-friendly
names.
Return a table with: Raw_Header, Suggested_Clean_Header, Reason.

Rules:
- Use clear names that banking users can understand.
- Avoid unclear short forms.

Dataset:
Raw_Header | Example_Value
Cust No | C1001
Txn Amt | 2500
Br | KL Main
Stat | Approved
KYC Stat | Completed
```

25. Create Error Flags Instead of Deleting

Simple explanation

In banking, suspicious records should often be flagged, not deleted.

Banking use case

Supports auditability, human review, and safer AI adoption.

Sample mini dataset

Record_ID	Amount	Account_No	Email
R251	950000	100045678901	ahmad@email.com
R252		ABC123	siti.gmail.com
R253	300	100045678903	lim@email.com

Trainer line

"Data cleaning is a business quality activity. For this technique, participants should focus on the risk of using the data without checking it."

READY-TO-COPY PARTICIPANT PROMPT

```
You are creating data quality flags for a banking operations dataset.

Do not delete any record. Create flags for records that need review.
Return a table with: Record_ID, Data_Quality_Flag, Issue_Found,
Risk_Level, Recommended_Human_Action.

Flag examples: Outlier Amount, Missing Amount, Invalid Account Number,
Invalid Email, Valid Record.

Dataset:
Record_ID | Amount | Account_No | Email
R251 | 950000 | 100045678901 | ahmad@email.com
R252 | | ABC123 | siti.gmail.com
R253 | 300 | 100045678903 | lim@email.com
```

Trainer Closing Summary

Clean → Validate → Mask → Review → Report

Theme	Message for participants
Completeness	Missing values should be handled carefully. Do not guess important banking data.
Consistency	Branch, product, city, and status values should use standard names.
Validity	Account numbers, emails, mobile numbers, dates, and amounts should follow business rules.
Safety	Sensitive customer data should be masked before training, sharing, or using public AI tools.
Auditability	Suspicious records should be flagged for review rather than silently deleted.
Human validation	AI can help identify issues, but banking staff must validate the final decision.

Final trainer line

Clean data leads to better MIS reports, better customer service, better compliance, safer AI adoption, and better business decisions.